# EXPERIMENT NO: 18

Experiment to understand KNN algorithm for a given dataset

## Program:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Iris.csv')
print(df.info())
print(df.variety.value_counts())
print(df.head())
features=df.iloc[:,:-1].values
label=df.iloc[:,4].values
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2,ran
dom_state=42)
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)
print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))
from sklearn.metrics import confusion_matrix
print(confusion_matrix(label,model_KNN.predict(features)))
from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

## Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    object
```

```
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None
variety
Setosa        50
Versicolor    50
Virginica     50
Name: count, dtype: int64
   sepal.length  sepal.width  petal.length  petal.width variety
0           5.1          3.5           1.4          0.2  Setosa
1           4.9          3.0           1.4          0.2  Setosa
2           4.7          3.2           1.3          0.2  Setosa
3           4.6          3.1           1.5          0.2  Setosa
4           5.0          3.6           1.4          0.2  Setosa
0.9666666666666667
1.0
[[50  0  0]
 [ 0 47  3]
 [ 0  1 49]]
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        50
  Versicolor       0.98      0.94      0.96        50
   Virginica       0.94      0.98      0.96        50

    accuracy                           0.97       150
   macro avg       0.97      0.97      0.97       150
weighted avg       0.97      0.97      0.97       150
```